

User Authentication & Localized Login Portal

1. INTERFACE OVERVIEW

The entry portal of the FarmFleet application provides a clean, localized landing and login experience designed specifically for agricultural users in the Coimbatore district. The design utilizes a high-contrast dark green gradient overlay with a central modern frosted-glass modal.

2. CORE FUNCTIONAL COMPONENTS

- **Branding & Identity Header:** Prominently features the FarmFleet logo along with explicit regional tagging (*FarmFleet Coimbatore: Agri-Rental & Marketplace*).
- **Credential Input Fields:**
 - **Full Name Input:** Custom text input field with placeholder formatting (e.g., *Ramesh Kumar*).
 - **Mobile Number Input:** Standard phone number validation field ensuring seamless OTP generation.
- **Authentication Action:** A high-visibility *Secure Login* → button triggering the backend authentication pipeline.

3. TECHNICAL ARCHITECTURE & UX STRATEGY

- **Design Pattern:** Glassmorphism overlay on top of field imagery, reducing visual strain for field workers operating devices in daylight.
- **Security Protocol:** Phone-number-first authentication workflow designed to minimize onboarding friction for rural farmers who may not utilize primary email accounts.

1. INTERFACE OVERVIEW

This screen serves as the primary discovery engine for farmers. It displays the broad machinery catalog anchored to the user's specific field location, offering unified search and dynamic distance calculations.

2. CORE FUNCTIONAL COMPONENTS

- **Location Header Bar:** Fixed bar displaying active location (My Field Location: Thondamuthur, Coimbatore) with fast-switching controls.
- **Category Filter Bar:** Horizontal chip selection including *All*, *Tractors*, *Harvesters*, *JCB*, *Agri Drones*, and *Borewell Rigs*.
- **Global Search Bar:** Real-time search supporting model names and tags (Search 'Farmtrac', 'Drone' ...).
- **Side-Panel Filter Controls:**
 - **Search Radius Slider:** Range selector configured to **100 km**.
 - **Operator Requirement:** Radio options for *Any*, *With Driver/Pilot*, and *Self-Drive*.

3. TECHNICAL ARCHITECTURE & LOCALIZATION

- **Total Volume Indicator:** Prominently displays aggregate regional supply (Showing 35 Vehicles near Thondamuthur).
- **Localization Control:** Embedded language switcher (தமிழ்) for instant Tamil language UI translation.
- **Proximity Logic:** Dynamically calculates radial distance markers (e.g., 18 km, 31 km, 55 km) from GPS coordinates.

1. INTERFACE OVERVIEW

A focused catalog view filtering the platform machinery specifically down to heavy-duty agricultural tractors available for rent in the target region.

2. CATEGORY METRICS & STANDARDIZED RATES

9 Vehicles

AVAILABLE NEAR LOCATION

₹950 / hr

STANDARD HOURLY RATE

3. DETAILED MODEL LISTINGS

Model Name	Base Location	Proximity	Status
Mahindra 575 DI	Sulur, TN	29 km	AVAILABLE
Massey 241	Mettupalayam, TN	15 km	AVAILABLE
Massey-Ferguson	Saravanampatti, TN	48 km	CURRENTLY BUSY
Sonalika 745	Singanallur, TN	32 km	AVAILABLE

4. OPERATIONAL LOGIC

Enforces standardized utility rates across tractor listings to prevent vendor price gouging while dynamically disabling checkout triggers for units flagged as Busy.

Marketplace Catalog — Harvester Fleet Directory

1. INTERFACE OVERVIEW

This view displays high-capacity harvesting equipment tailored for seasonal grain and crop harvesting cycles across Tamil Nadu agricultural belts.

2. CATEGORY METRICS & ACREAGE BILLING

5 Vehicles

HARVESTER FLEET COUNT

₹2,400 / acre

AREA BILLING RATE

3. MACHINERY LISTINGS

Harvester Model	Base Location	Distance	Status
Claas Tiger	Gandhipuram, TN	56 km	BUSY
Kartar 4000	Peelamedu, TN	26 km	AVAILABLE
Kubota DC-68 G	Pollachi, TN	23 km	AVAILABLE
Mahendra Harvester	Annur, TN	42 km	BUSY

4. TECHNICAL UNIT ECONOMICS

Shifts calculation metrics from time-based (hourly) to area-based (acreage) to accurately reflect standard agricultural operational billing for harvesters.

Marketplace Catalog — Earthmovers & JCB Fleet

1. INTERFACE OVERVIEW

Displays heavy earthmoving and land-preparation machinery essential for farm leveling, canal digging, and field infrastructure maintenance.

2. CATEGORY METRICS & HOURLY STRUCTURE

3 Vehicles

EARTHMOVER FLEET

₹1,000 / hr

DRIVER + FUEL INCLUDED

3. MACHINERY LISTINGS

Equipment Name	Base Location	Distance	Availability
CAT 424	Gandhipuram, TN	27 km	CURRENTLY BUSY
Hitachi EX200	Peelamedu, TN	15 km	AVAILABLE
JCB 3DX	Pollachi, TN	39 km	AVAILABLE

4. VISUAL VERIFICATION & BUNDLING

Integrates rich imagery of physical assets to assist farmers in verifying machinery scale before dispatch while bundling operator and fuel costs directly.

Marketplace Catalog — Precision Agri Drones

1. INTERFACE OVERVIEW

Displays advanced drone technology listings used for modern precision agriculture tasks like liquid fertilizer spraying, pesticide deployment, and aerial mapping.

2. CATEGORY METRICS & SERVICE BUNDLING

₹1,100 / acre

PRECISION RATE

Pilot Included

CERTIFIED OPERATOR BADGE

3. DRONE FLEET LISTINGS

Drone Identifier	Regional Base	Distance	Status
Agri Drone 1	Gandhipuram, TN	41 km	BUSY
Agri Drone 2	Peelamedu, TN	46 km	AVAILABLE
Agri Drone 3	Pollachi, TN	14 km	AVAILABLE
Agri Drone 4	Annur, TN	43 km	BUSY

4. RISK MANAGEMENT STRATEGY

Drone rentals automatically mandate certified pilots to eliminate operational liability and equipment damage risks for untrained farmers.

Marketplace Catalog — Heavy Borewell Drilling Rigs

1. INTERFACE OVERVIEW

Displays specialized deep-drilling vehicles utilized for farm irrigation and groundwater extraction across varying soil depths and geological formations.

2. DRILLING DEPTH PARAMETERIZATION

Machine capabilities are indexed primarily by maximum drilling depth capacity rather than traditional horsepower metrics:

Rig Designation	Regional Base	Max Depth Capacity	Status
Borewell Rig 1	Gandhipuram, TN	600 Ft Max	BUSY
Borewell Rig 2	Peelamedu, TN	650 Ft Max	AVAILABLE
Borewell Rig 3	Pollachi, TN	17 km base (700 Ft)	AVAILABLE
Borewell Rig 4	Annur, TN	750 Ft Max	BUSY
Borewell Rig 7	Saravanampatti, TN	900 Ft Max	BUSY
Borewell Rig 8	Singanallur, TN	950 Ft Max	AVAILABLE
Borewell Rig 9	Ukkadam, TN	1000 Ft Max	AVAILABLE

3. DYNAMIC SPECIFICATION SWAPPING

The UI frontend dynamically swaps vehicle attribute labels to match specialized drilling industry parameters (depth in feet).

Security & Identity Verification (KYC Modal)

1. INTERFACE OVERVIEW

A security modal overlay designed to authenticate farmer identity prior to finalizing high-value machinery rental agreements.

2. CORE FUNCTIONAL COMPONENTS

- **Modal Focus Overlay:** Blurred background masking preventing interaction with background elements during verification.
- **Item Context Card:** Displays active booking target (*Farmtrac 3630 TX Plus | Tractors | Driver Included*).
- **Form Fields:**
 - **Aadhaar Number Field:** Formatted input holding 12-digit UID (7789456123741).
 - **OTP PIN Field:** Masked 4-digit passcode entry (••••) with demo test indicator (USE 1234 FOR DEMO).
- **Actions:** Secondary Back trigger and primary Verify & Proceed button.

3. SECURITY & LIABILITY ARCHITECTURE

Mandating identity checks links physical equipment dispatches directly to government UID records, safeguarding valuable fleet assets against loss or misuse.

Payment Gateway & Flexible Settlement Modal

1. INTERFACE OVERVIEW

An interactive financial checkout layer offering flexible payment arrangements tailored to agricultural cash-flow realities.

2. SETTLEMENT STRUCTURE

₹4,450

GRAND TOTAL AMOUNT

₹890

20% ADVANCE OPTION

3. PAYMENT TIER SELECTIONS

- **Option A — Full Payment:** Primary green action button (Pay Full Amount Now – ₹4,450).
- **Option B — Deferred Advance:** Secondary outlined button (Pay 20% Advance (Balance on completion) – ₹890).

4. FINANCIAL INCLUSION UX STRATEGY

Reduces upfront capital constraints for smallholder farmers by requiring only a minor deposit to lock in machinery bookings prior to work completion.

Active Booking Tracking Overlay

1. INTERFACE OVERVIEW

A modal window giving farmers instant visibility over active rental orders, unique order identifiers, and payment fulfillment states.

2. ACTIVE ORDER CARD METADATA

Attribute	System Data Value
Assigned Equipment	Farmtrac 3630 TX Plus
Order Tracking ID	Order ID: B1785573826360
Equipment Owner	Kovai Fleet Works
Payment Ledger	Paid: ₹4,450 / Total: ₹4,450
Operational Status	ACTIVE

3. REAL-TIME ORDER MANAGEMENT

Enables farmers to monitor dispatch state and verify completed transactions across multi-vehicle active rentals from a central interface.

1. INTERFACE OVERVIEW

The main administrative hub for equipment owners, presenting high-level business analytics, fleet metrics, and live field assignments.

2. BUSINESS PERFORMANCE INDICATORS

₹1,85,000

MONTHLY GROSS EARNINGS

40 Units

TOTAL REGISTERED FLEET

16 Jobs

ACTIVE FIELD JOBS

Madhuvarun

OWNER ACCOUNT

3. NAVIGATION & ONBOARDING TRIGGERS

- **Sidebar Menu:** Links for *Dashboard*, *Fleet Directory*, *Live Map*, and *Switch Workspace*.
- **Inventory Expansion:** Prominent + List Equipment action trigger.

Fleet Owner Inventory & Maintenance Controls

1. INTERFACE OVERVIEW

An operational management directory allowing equipment owners to alter machine maintenance states, monitor client allocations, and control rental inventory.

2. LIVE INVENTORY TABLE

Equipment Details	Status & Actions	Active Customer	Rate
Farmtrac 50 Gandhipuram, TN	BROKEN Action: MARK REPAIRED	No active client	₹950 / hr
Farmtrac 3630 TX Plus Peelamedu, TN	BOOKED Action: MARK BROKEN	Raju F. (Active)	₹950 / hr
John Deere 5310 Pollachi, TN	BOOKED	Raju F. (Active)	₹950 / hr
Kubota MU4501 Annur, TN	BOOKED	Raju F. (Active)	₹950 / hr
Mahindra 575 DI Sulur, TN	IDLE	No active client	₹950 / hr

3. MAINTENANCE GOVERNANCE

Provides one-click toggles (MARK BROKEN / MARK REPAIRED) so owners can immediately remove damaged units from search results.

System Administrator Governance & Onboarding Console

1. INTERFACE OVERVIEW

The master admin console used by platform operators to monitor platform analytics, handle disputes, and verify vendor onboarding paperwork.

2. MASTER PLATFORM INDICATORS

₹1,85,000

TOTAL REVENUE

16 Live

ACTIVE RENTALS

6 Urgent

PENDING APPROVALS

0 Disputes

ALL SYSTEMS OPERATIONAL

3. ONBOARDING VERIFICATION QUEUE

Owner	Equipment Type	Location	Docs	Status
Suresh Kovai	Farmtrac 50	Gandhipuram, TN	✓	IN PROGRESS
Kovai Fleet Works	Farmtrac 3630 TX Plus	Peelamedu, TN	✓	VERIFIED
Velan Agri Services	John Deere 5310	Pollachi, TN	✓	VERIFIED
Kongu Farm Rentals	Kubota MU4501	Annur, TN	✓	IN PROGRESS